

Banking Customer Analytics & Fraud Detection Report

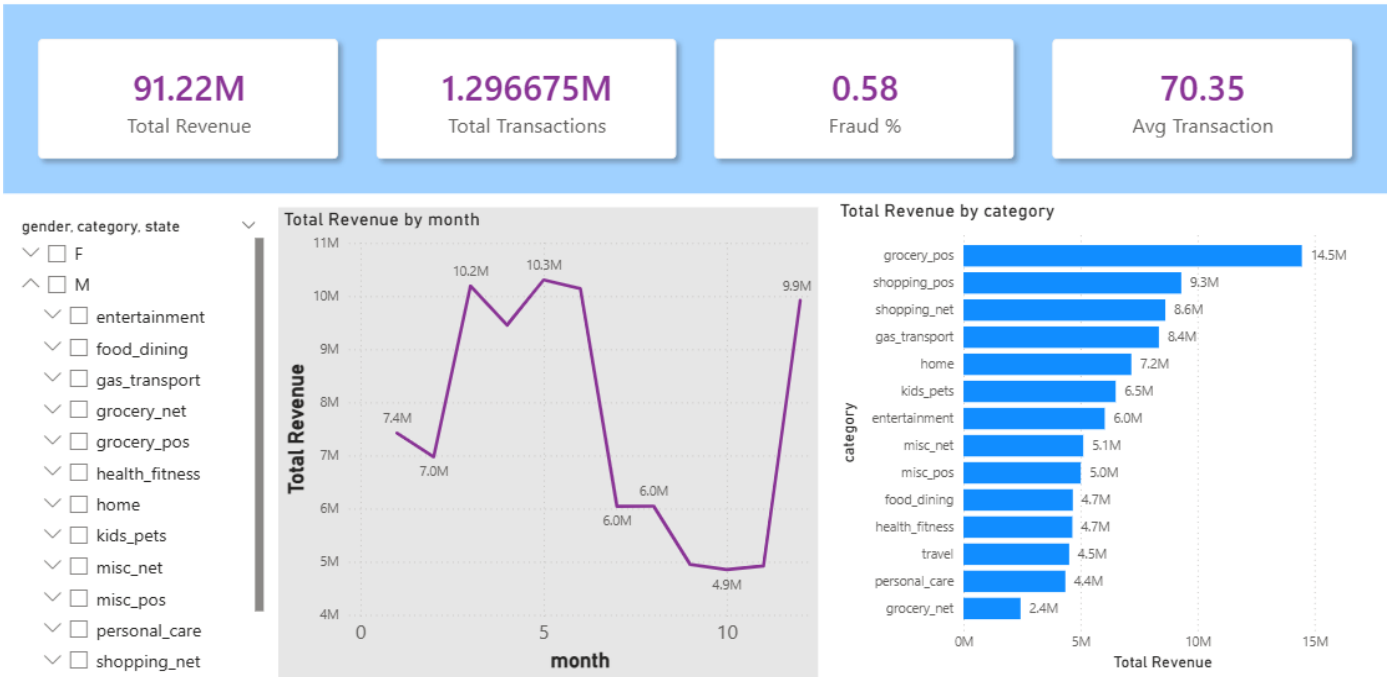
This report presents an end-to-end Banking Customer Analytics and Fraud Detection System developed using MySQL, Python, Machine Learning, and Power BI. The project analyzes large-scale banking transaction data to identify fraud trends, customer spending behavior, revenue patterns, and business risks. The dashboards were designed to support executive decision-making, operational monitoring, and fraud risk management in a consumer banking environment.

Executive Summary

- Total Revenue generated across transactions exceeded 91M.
- The system processed over 1.29M transactions.
- Fraud percentage observed was approximately 0.58%.
- Grocery and shopping categories contributed the highest transaction revenue.
- Fraud activities were concentrated in grocery_pos, shopping_net, and misc_net categories.
- Late-night hours showed significantly higher fraud activity patterns.
- Middle-aged customers contributed the highest revenue share among all age groups.

Executive Overview Dashboard

BANKING ANALYTICS: TRANSACTION & FRAUD PERFORMANCE REPORT

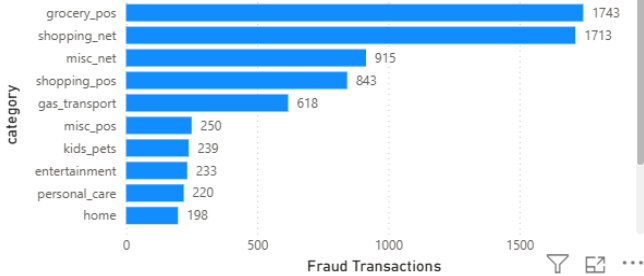


Fraud Analytics Dashboard

FRAUD ANALYTICS

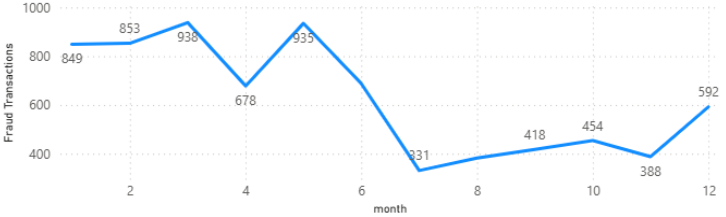
FRAUD BY CATEGORY

Fraud Transactions by category

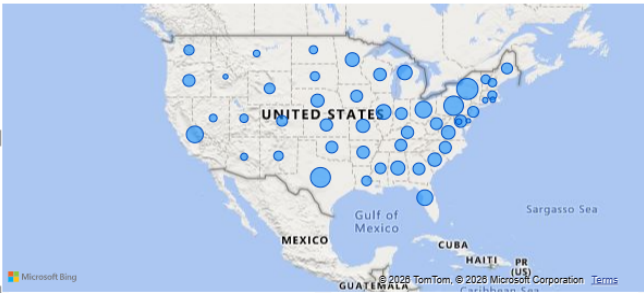


FRAUD TREND OVER TIME

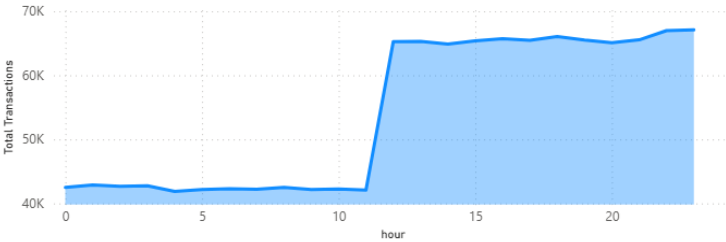
Fraud Transactions by month



Fraud Transactions by state



Total Transactions by hour



Fraud increases during late-night hours.

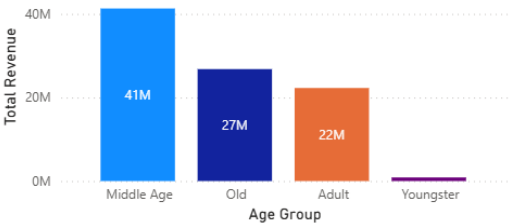
Customer Insights Dashboard

CUSTOMER INSIGHTS

AGE GROUP SPENDING

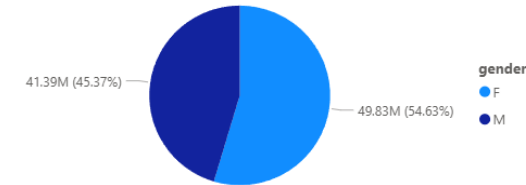
Total Revenue by Age Group and Age Group

Age Group ● Middle Age ● Old ● Adult ● Youngster



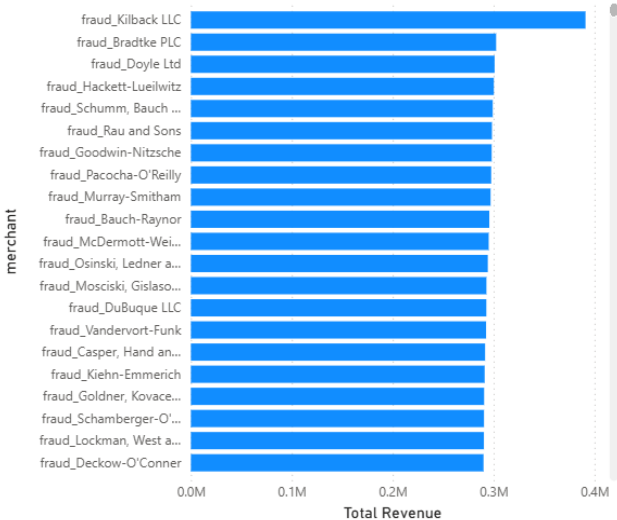
TOP MERCHANTS

Total Revenue by gender



TOP MERCHANTS

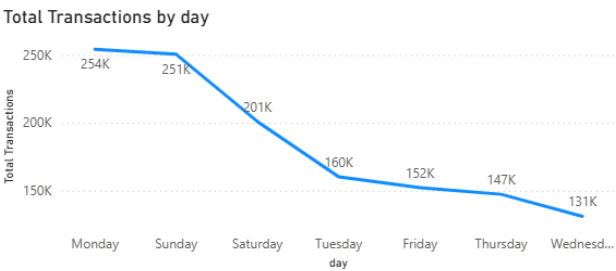
Total Revenue by merchant



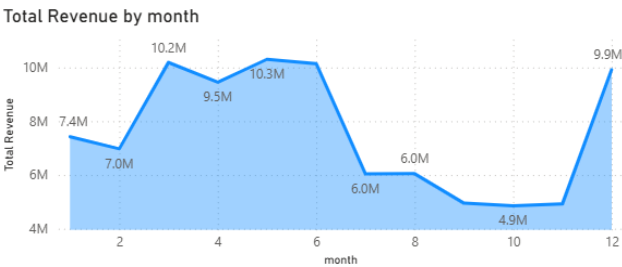
Revenue Trends Dashboard

REVENUE TRENDS

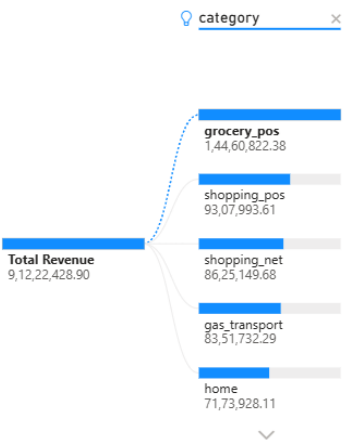
DAILY TRANSACTION TREND



MONTHLY REVENUE



CATEGORY REVENUE



Detailed Business Insights

- Revenue Performance:** Grocery POS and shopping-related categories generated the highest transaction revenue, indicating strong consumer dependence on retail and daily essentials.
- Fraud Concentration:** Fraudulent activities were heavily concentrated in online shopping and grocery POS categories, suggesting vulnerabilities in high-frequency consumer transaction channels.
- Fraud Time Analysis:** Fraud activity increased significantly during late-night hours, indicating possible automated or suspicious transaction behavior patterns.
- Customer Demographics:** Middle-aged customers contributed the largest share of total revenue, making them the most valuable customer segment.
- Geographical Fraud Analysis:** Fraud incidents were distributed across multiple states, indicating the need for region-based fraud monitoring systems.
- Merchant Analysis:** Certain merchants demonstrated disproportionately high transaction volumes, indicating opportunities for merchant risk scoring and transaction monitoring.
- Transaction Trends:** Monthly revenue fluctuations suggest seasonal spending behavior and changing customer purchasing patterns.

Potential Risks Identified

- Increasing online transaction fraud may result in financial losses and reduced customer trust.

- Late-night fraud patterns indicate potential system abuse and require real-time fraud monitoring mechanisms.
- Concentration of revenue within limited customer segments may increase business dependency risk.
- High transaction categories require stricter fraud validation and anomaly detection systems.
- Geographically distributed fraud patterns indicate the need for location-based authentication models.

Strategic Recommendations

- Implement real-time fraud detection systems using machine learning models.
- Enable transaction anomaly alerts for unusual late-night spending behavior.
- Introduce customer risk profiling and dynamic fraud scoring mechanisms.
- Deploy geo-location validation and merchant trust scoring systems.
- Build automated reporting pipelines for executive and operational monitoring.
- Enhance customer segmentation strategies for targeted financial services and personalized offers.

Technical Implementation

Component	Technology Used
Database	MySQL
Programming Language	Python
Data Analysis	Pandas, NumPy
Machine Learning	Scikit-learn
Visualization	Power BI
Dashboarding	DAX Measures & Interactive Visualizations
Data Pipeline	Python ETL Workflows

Conclusion

The Banking Customer Analytics & Fraud Detection System demonstrates strong analytical, reporting, and business intelligence capabilities aligned with enterprise banking environments. The project highlights expertise in SQL, Python, Power BI, dashboard development, fraud analytics, customer segmentation, and executive reporting. The solution showcases the ability to solve unstructured business problems, generate actionable insights, and support strategic decision-making through data-driven approaches.